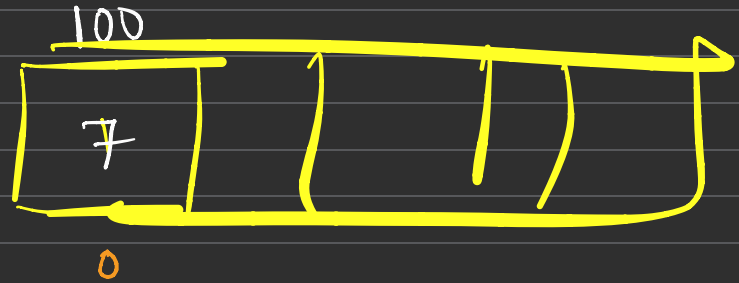


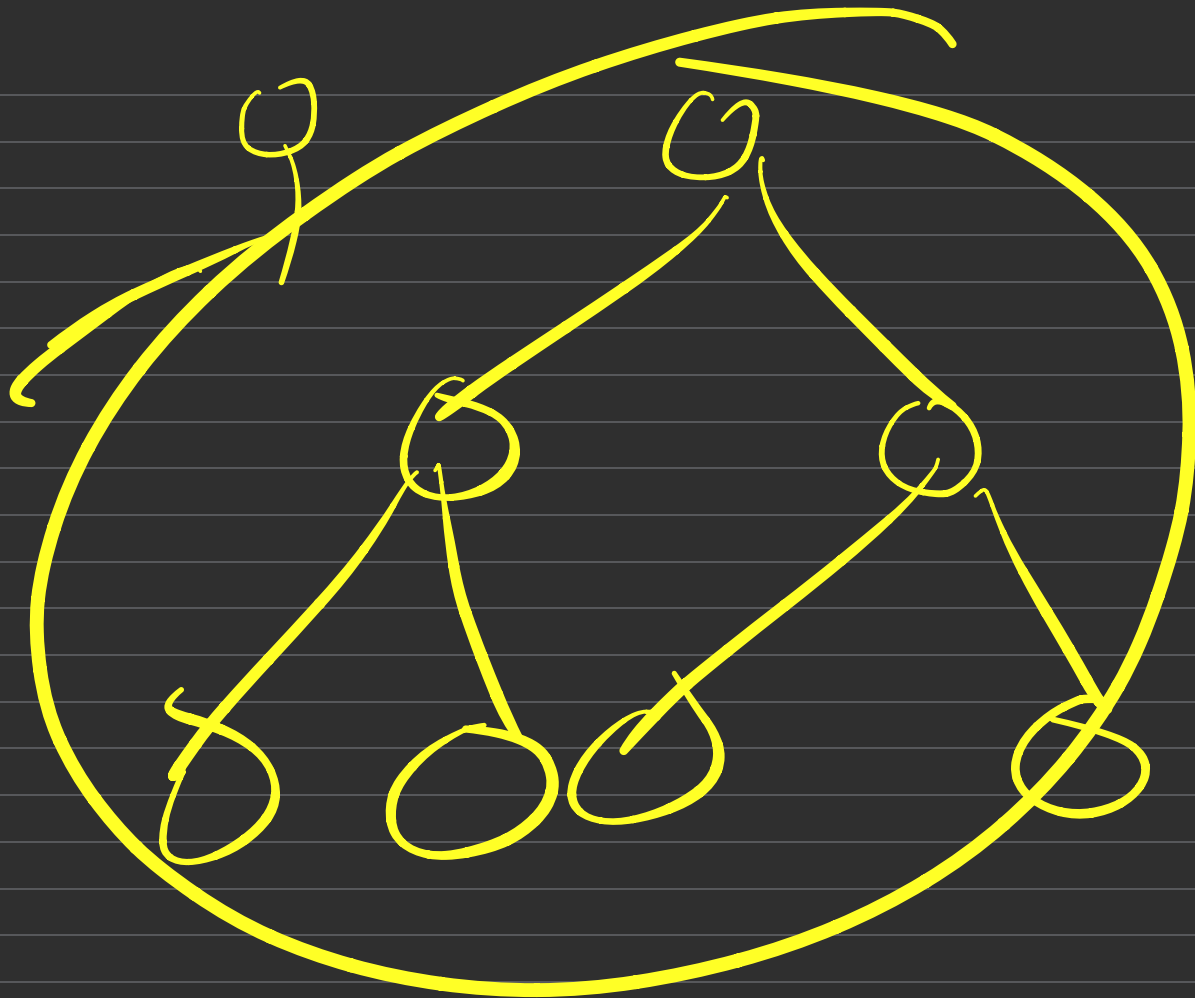
`int arr() = { 7, 4, 8, 9 }`

`print ( arr[3] )` $\rightarrow 9$



$\left\{ \begin{array}{l} * (arr + 3) \\ * (100 + 3 \times 4) \\ * (100 + 12) \\ * (\underline{12}) \end{array} \right.$

9



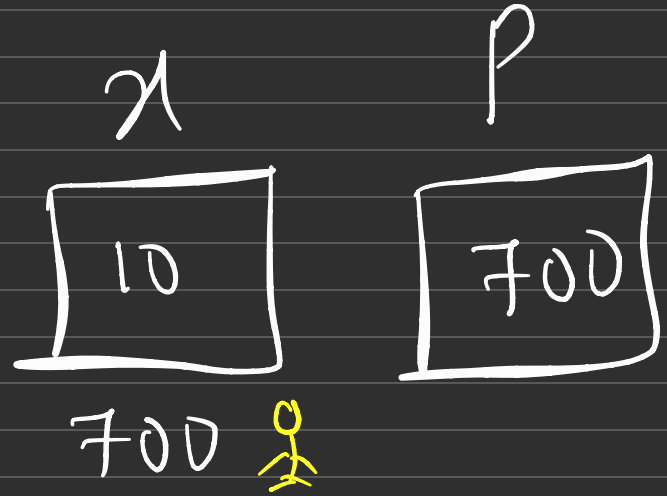
```
int x = 10
```

```
int *p = &x
```

```
print(x) = 10
```

```
print(p) = 700
```

```
print(*p) = 10
```



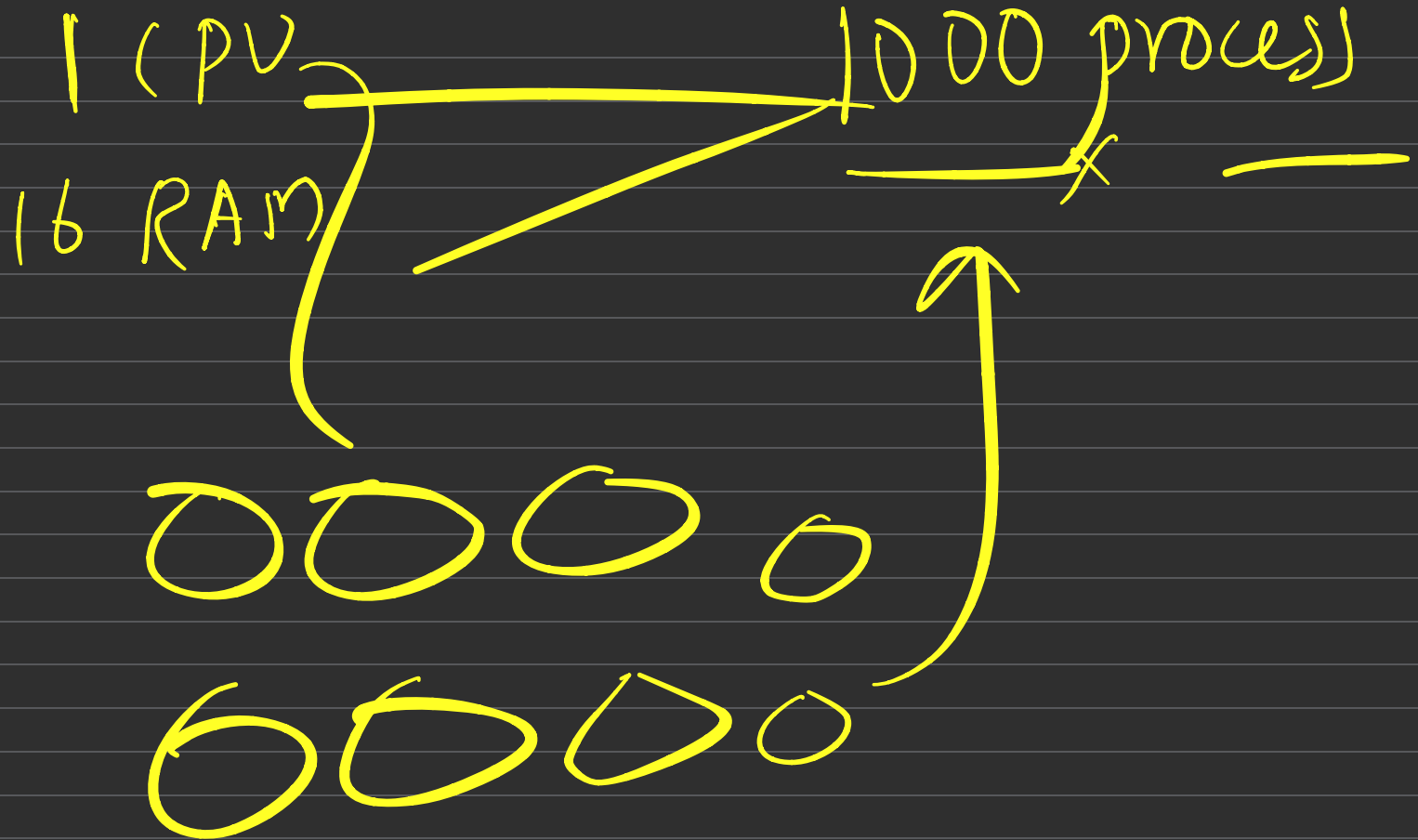
int α

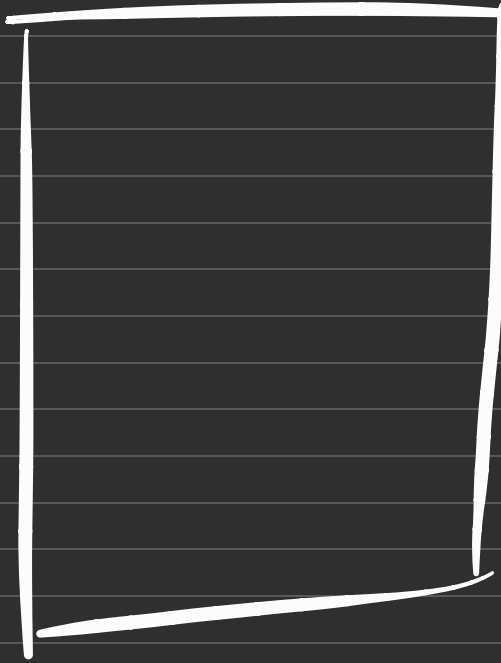
1

α
10

bul

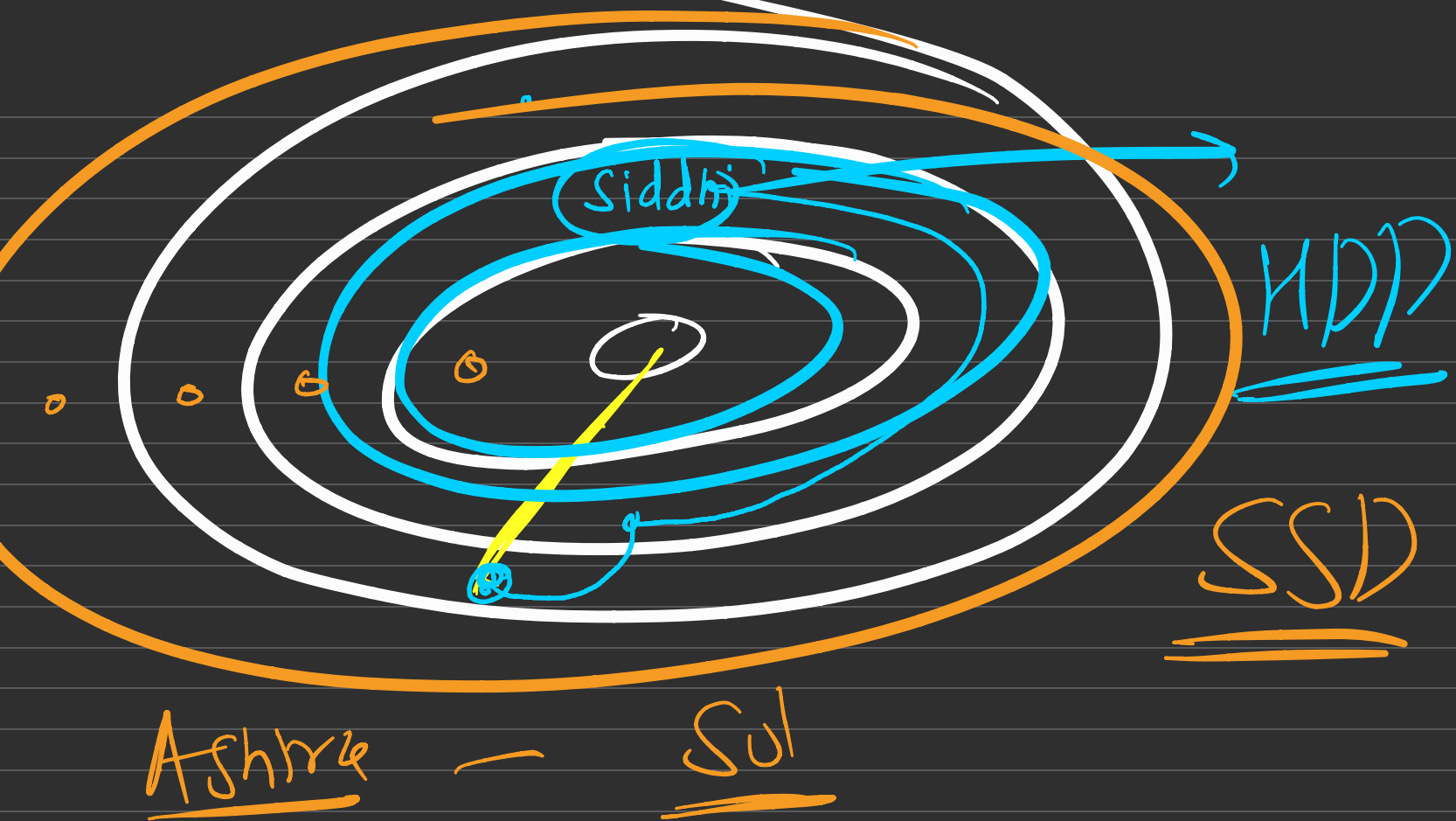
fr



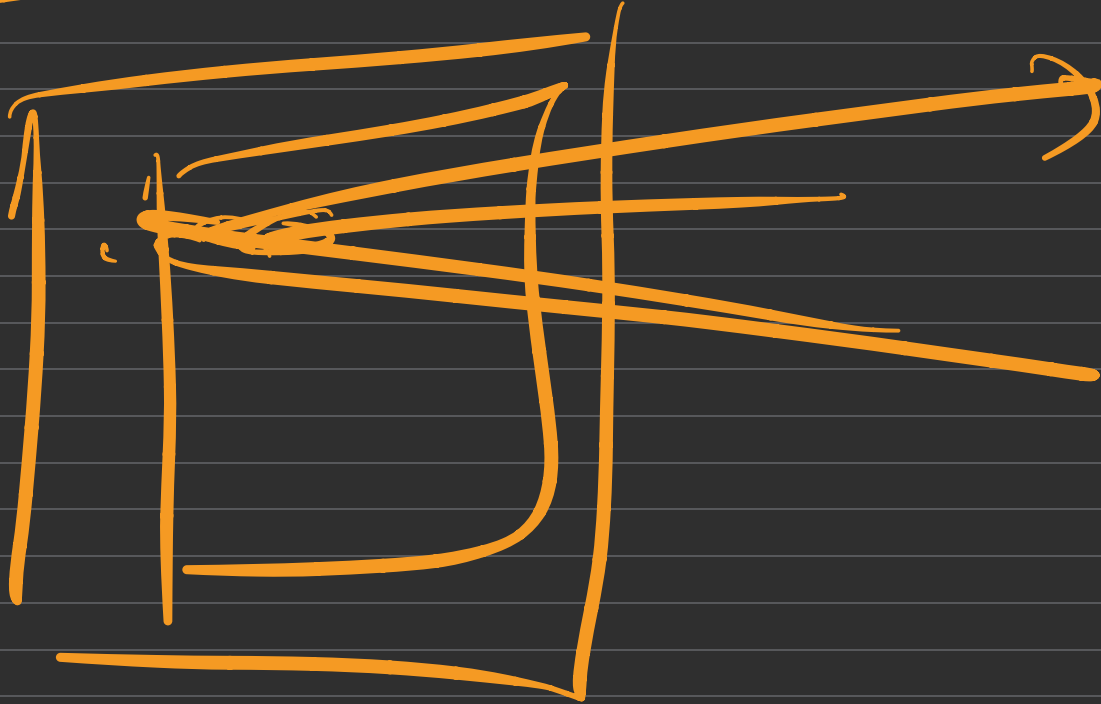


Capacitor

└ decharge if
you stop power
supply



SSD — solid state drive





WELCOME TO JAVA PROGRAMMING



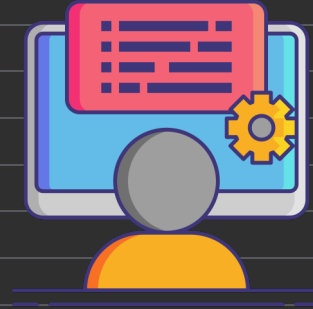
“Don’t wait for the perfect moment—take the moment and make it perfect. Every small step you take today is building the life you’ll be proud of tomorrow.” ✨

- Rules → protocol
- instruction

What is Program

A program is a set of instructions given to a computer to perform a specific task.

1. Take first number
2. Take second number
3. Add them
4. Display the result



What is Program Language

A program is a set of instructions given to a computer to perform a specific task in a specific language.



Enterprise
level
Application

PROGRAMMING LANGUAGES AND THEIR USES

PYTHON

- 1) Data Science
- 2) Machine Learning
- 3) Web Development
- 4) Automation
- 5) Game Development
- 6) Data analysis
- 7) Data visualization
- 8) Artificial intelligence

JAVA

- 1) Android Apps
- 2) Server-Side Apps
- 3) Enterprise Apps
- 4) Web Based Apps
- 5) Big data
- 6) Game Development
- 7) Internet of things
- 8) Cloud computing

C++

- 1) Games Development
- 2) GUI Apps
- 3) OS
- 4) Database Systems
- 5) Embedded
- 6) Networking
- 7) Virtual Reality
- 8) Computer Vision

JAVASCRIPT

- 1) Server-side Dev
- 2) Web Dev and Apps
- 3) Mobile Apps
- 4) Machine Learning
- 5) IoT
- 6) Automation
- 7) Embedded system
- 8) Chatbot Development

SWIFT

- 1) IOS App Dev
- 2) Deep Learning
- 3) IOT
- 4) Server-side Dev
- 5) Open-source Dev
- 6) MacOS App Dev
- 7) Machine Learning
- 8) Automation

C#

- 1) Games Development
- 2) Web Dev and Apps
- 3) IOT
- 4) Backend Services
- 5) Windows App Dev
- 6) Robotics
- 7) Cloud computing
- 8) Database program

Application
X

C/C++

Java

↘

Python

Taxa 21

— JP | Bar | ms

25
=

What is Java and why is it used?

Java is a high-level, object-oriented programming language used to build software applications.

↓
human

low-level → Hindi/English (01)

Yes Why

No Why

Features of JAVA Programming

Portable

JAVA program compiled on one machine can run on other machine. JAVA is platform Independent.

Memory Management

Memory management in Java is automatically handled by the Java Virtual Machine (JVM).

Scalable

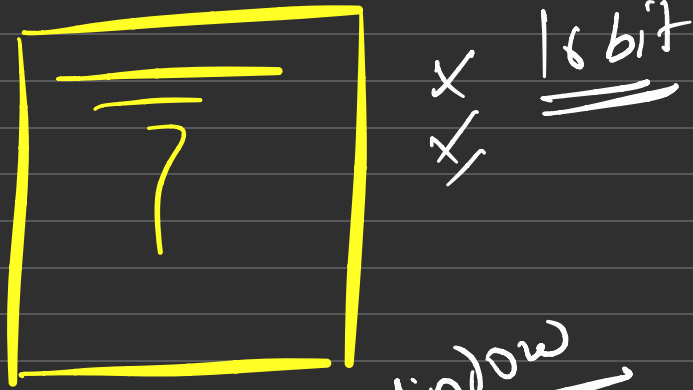
Java can handle both small and large-scale applications

Interpreted

Java code is not directly executed by the computer. It is first compiled into bytecode. This byte code is then understood by the JVM. This enables Java to run on any platform without rewriting code.

C/C++ → Platform

32 bits



.cpp
shradha } xxx

Window

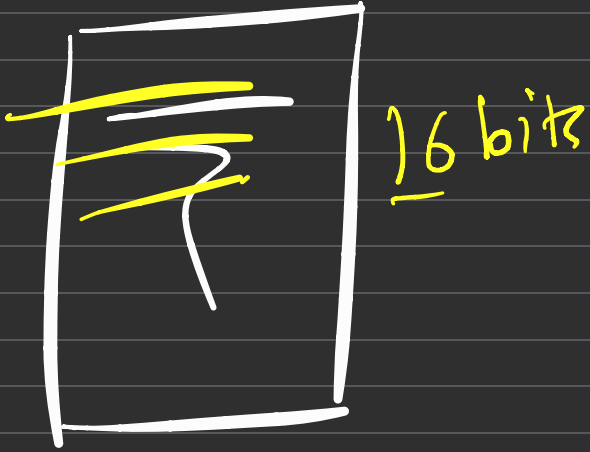
After Compile

.exe

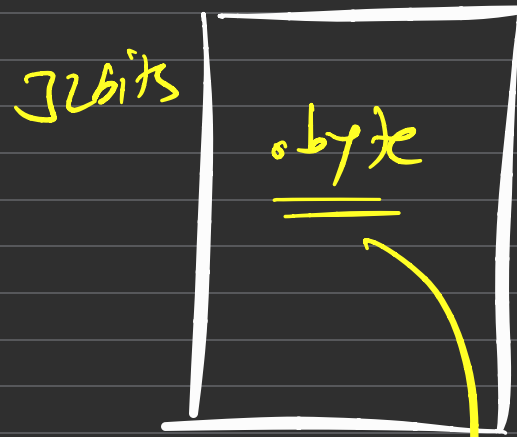
→ PID

Linux
.out

JVM / Compiler



JVM / Compiler



Window
Java xxx

Linux

compile

.byte

.exe

Run

new
malloc
calloc

Runtime

C/C++

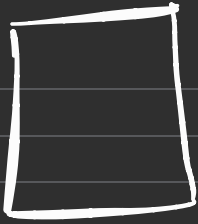
program

Memory
Leakage

3000

40/45

delete

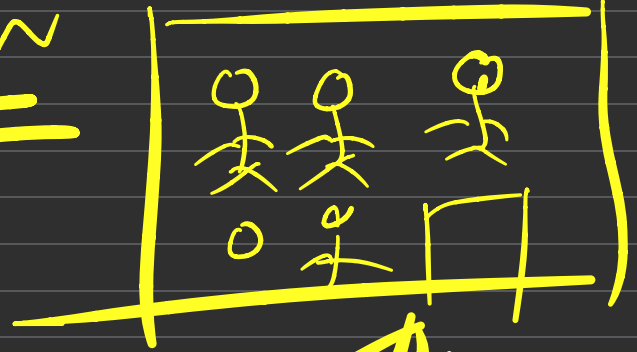


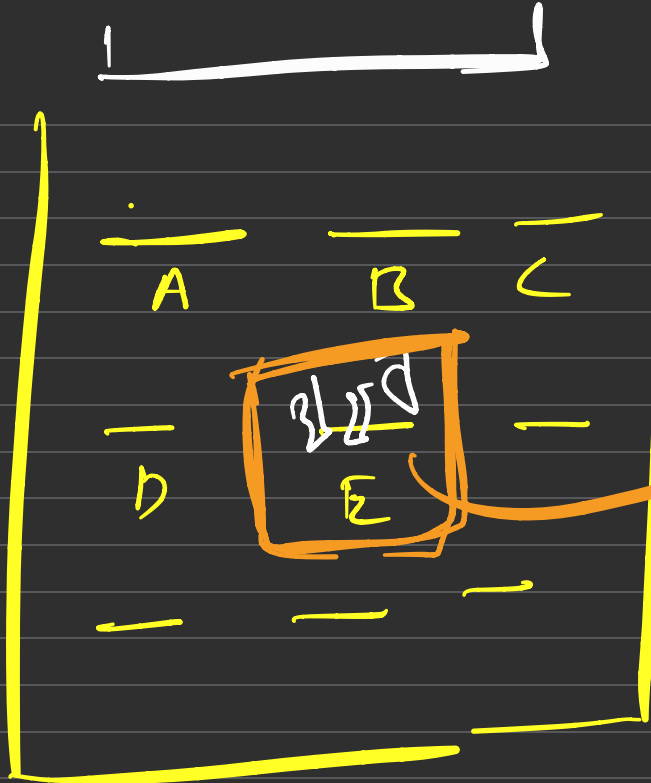
250 User



1000 User

300 Req

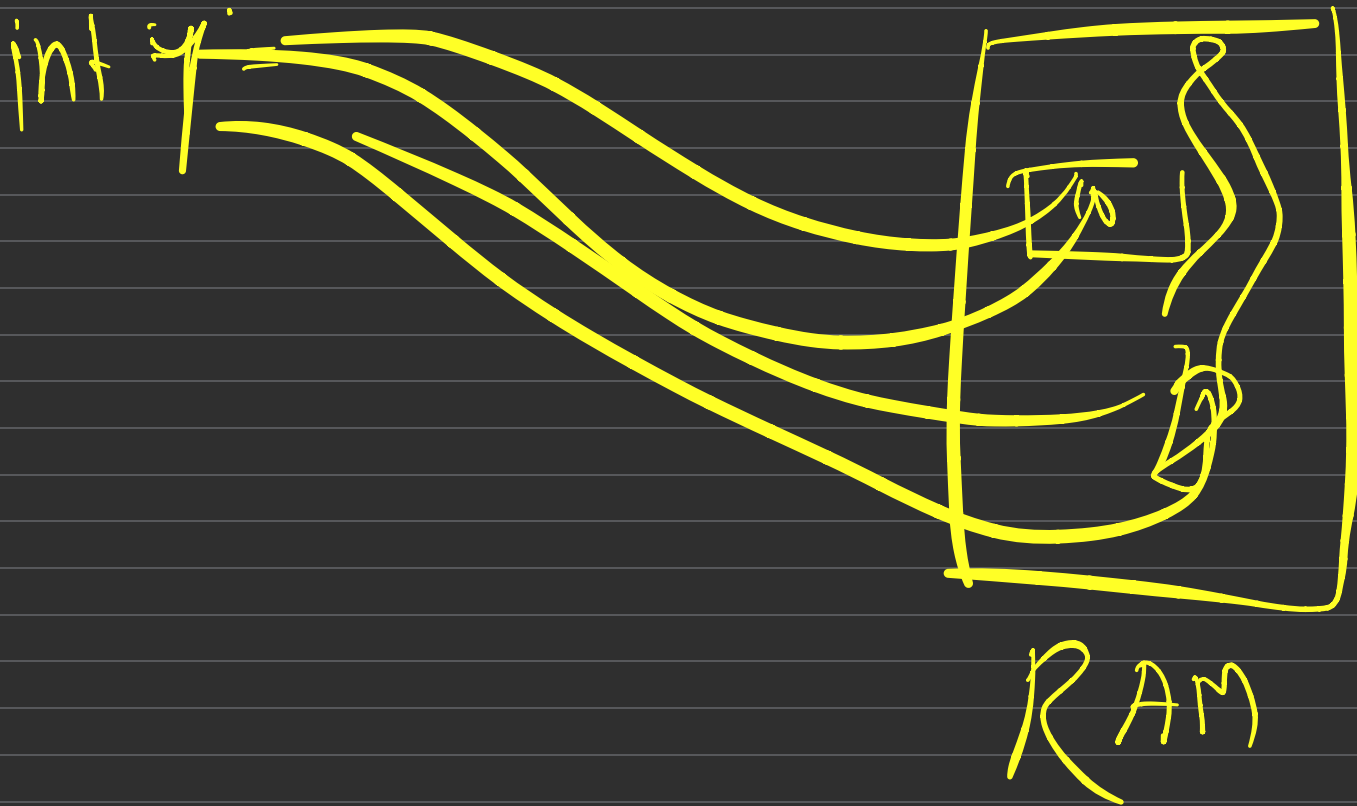


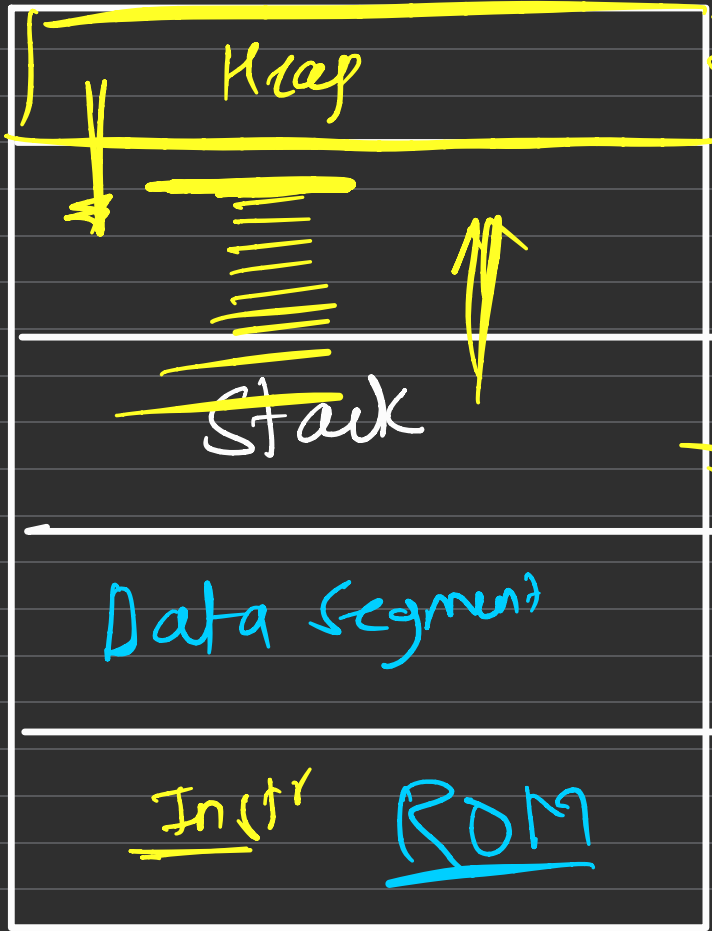


Sammundi

Garbage







Runtime (new/malloc call
printf("Hello world"))

Static

Variable
function

Compiler Vs Interpreter

Compiler

1. Mainly converts High level lang. into machine language so that computer can understand.
2. Compiles whole code at sometime.
3. CPU Utilisation Is more in compiler since whole code get compiled at a time.
4. C, C++, C#, etc are programming languages that are compiler-based.

Interpreter

1. Mainly converts High level lang. into intermediate language.
2. It translate the code line by line .
3. CPU utilisation is less in interpreter as code is translated line by line

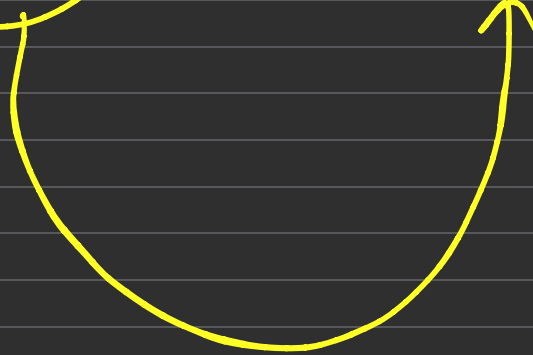
Python, Ruby, Perl, SNOBOL, MATLAB, etc are programming languages that are interpreter-based.

Compiler (C | C++)

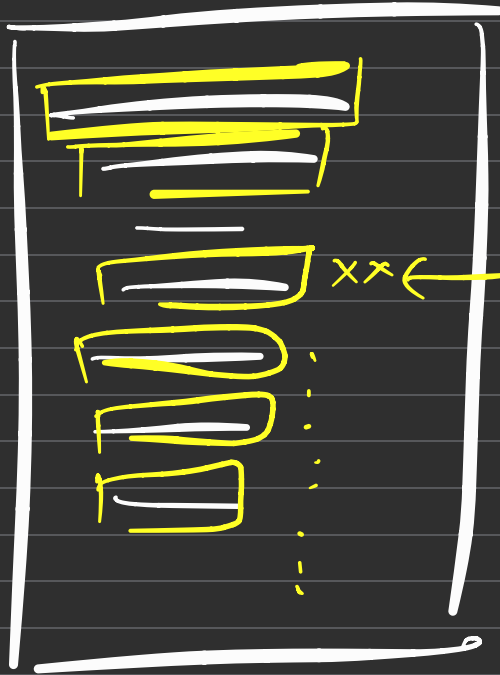


100 lines

(low level)
Binary lang



interpreter



100 line

Compile



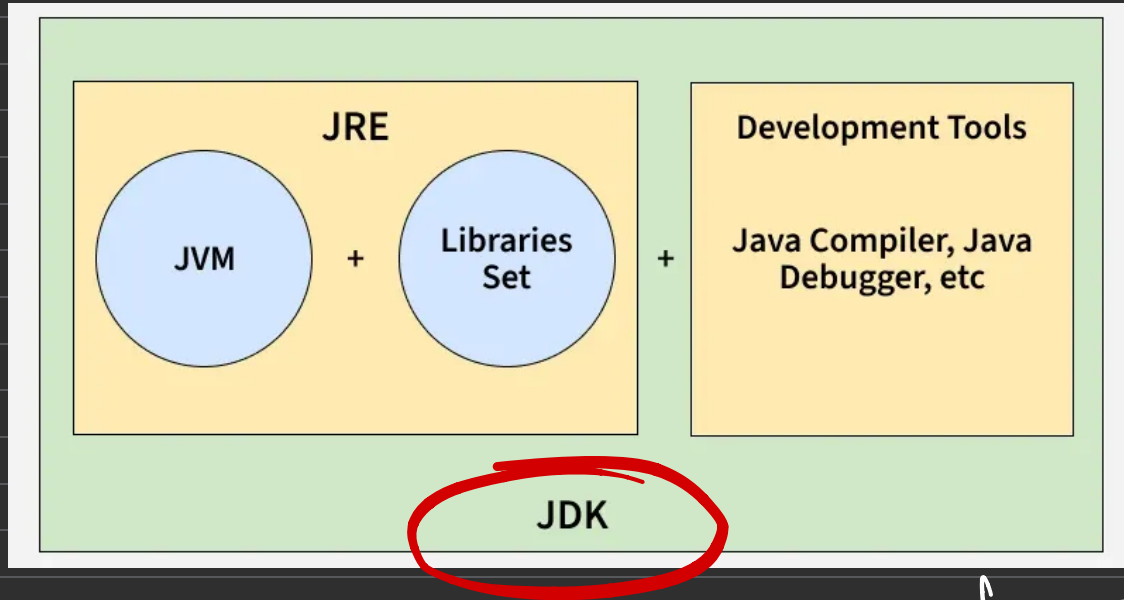
(intermediate)
Compressed
code



JDK JVM JRE

grt
gcc

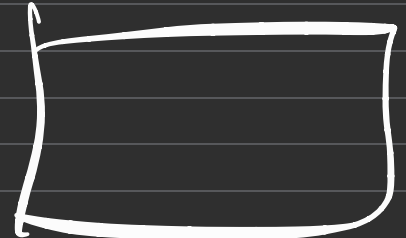
java --version



win + R

-

cmd



JDK → Java Develop. Kit
compiler, interpreter, libraries

Security

JRE - Java Runtime Env.

JVM — call main

```
int add()
```

```
}
```

```
{
```

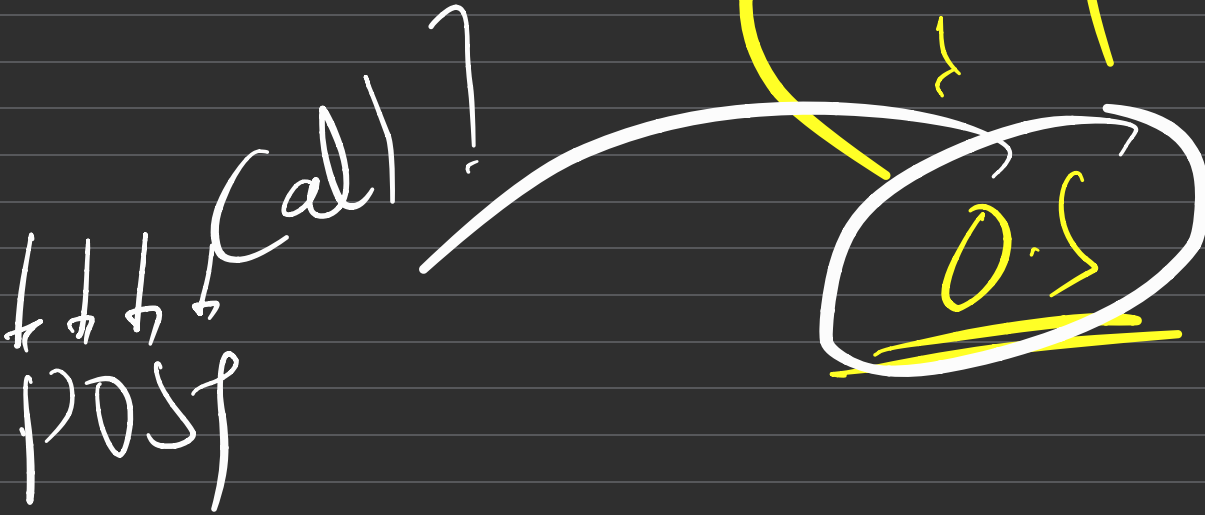
↑

```
int main()
```

```
}
```

```
{
```

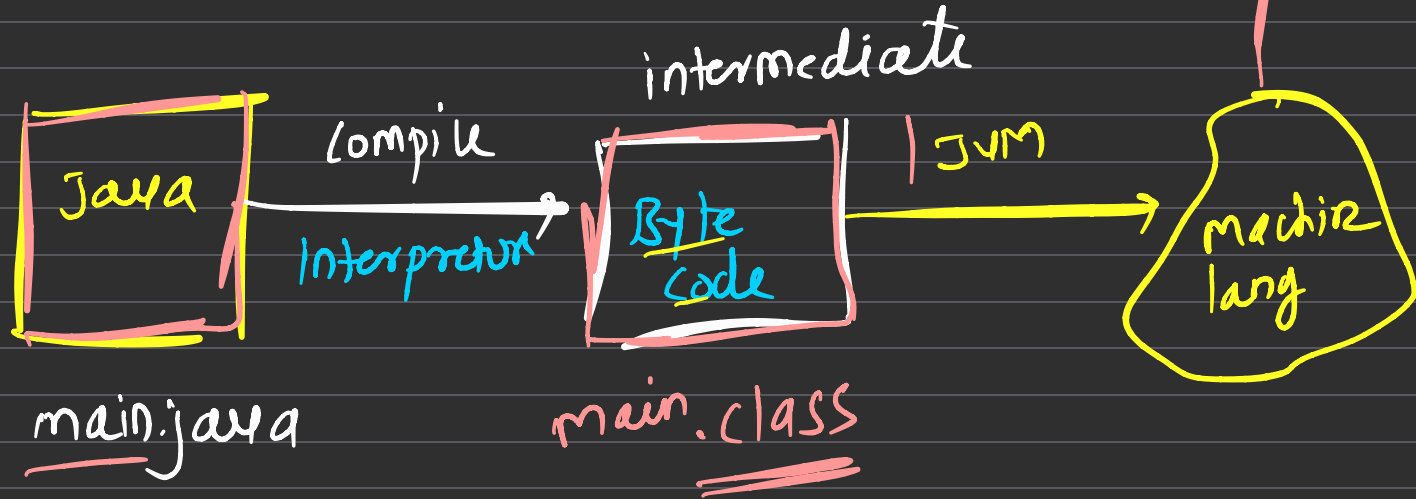
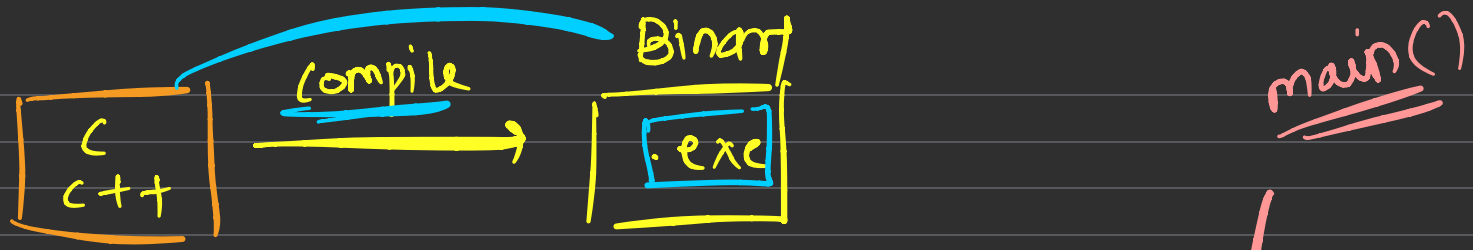
{ add() }



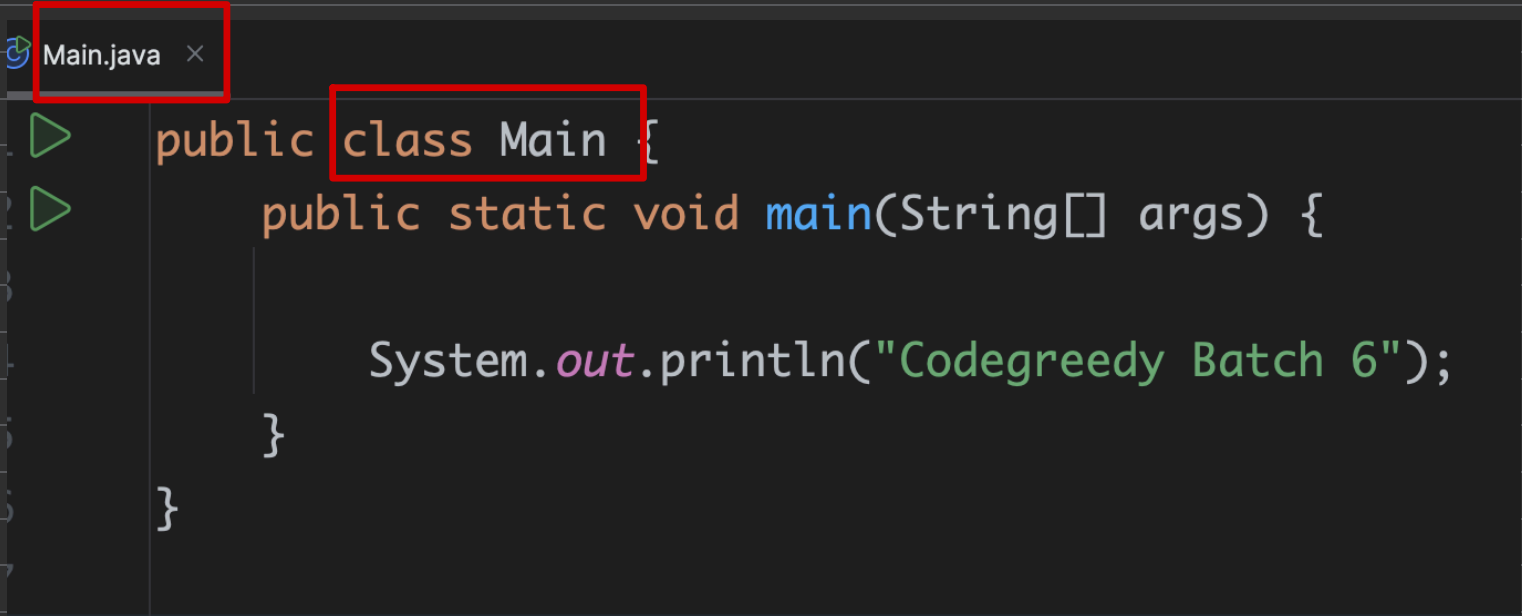
Source code → bytecode → JVM → execution.

Main.java ×

```
public class Main {  
    public static void main(String[] args) {  
  
        System.out.println("Codegreedy Batch 6");  
    }  
}
```



Why class name containing main function & File name must be same



The screenshot shows a code editor with a dark background. At the top left, a tab labeled "Main.java" with a close button is highlighted with a red rectangle. Below the tab, the code is displayed with syntax highlighting. The first line is "public class Main {" where "class Main" is highlighted with a red rectangle. The second line is "public static void main(String[] args) {" followed by an indented line "System.out.println(\"Codegreedy Batch 6\");" and then closing braces "}" and "}" on subsequent lines. Green play button icons are visible on the left margin of the first two lines of code.

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Codegreedy Batch 6");  
    }  
}
```

1.7 file = Main.java } Success
public class Main

file = Main.java } Error
public class Pramod

file = Main.java } Success
class Pramod

* Program start from $\longrightarrow$ main method
(jump)

class A

compile

A.class

class B



B.class

class C

C.class

public class Main

}

main()

{

main.class



Red



S

G

Yellow



K

B



A



V

```
public class Demo
```

```
    {  
        main
```

```
    }  
}
```

```
public class Adnan
```

```
    {  
        main()
```

```
    }  
}
```

```
public class Box
```

```
    {  
        main()
```

```
    }  
}
```

Main.java

java

public class Main

main()

Which Text Editor or IDE to use

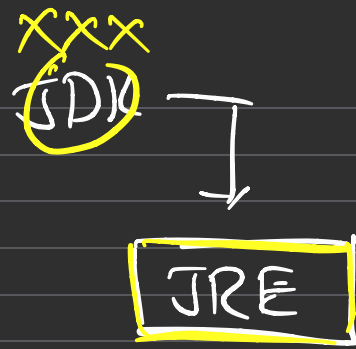
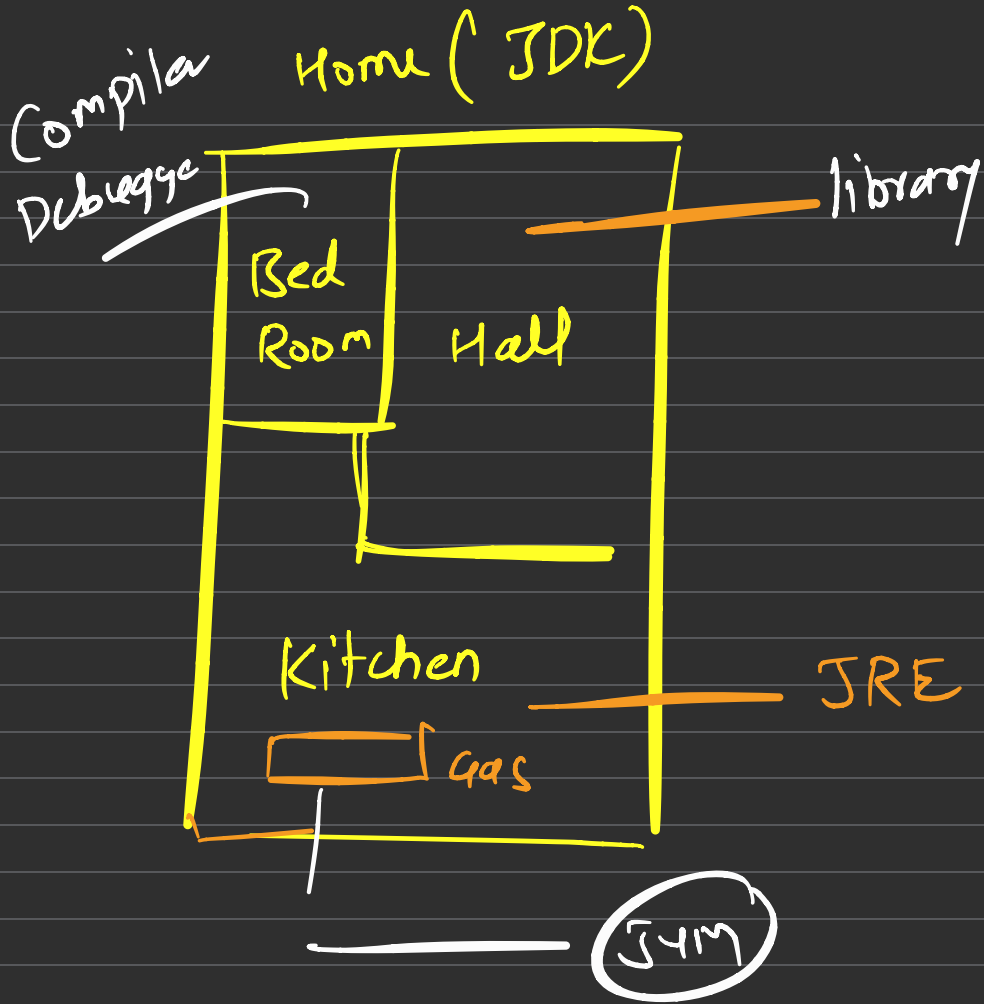
└ notepad++
└ sublime text

Command to Run java program on terminal

Go to the path/folder - in which your java program is saved.

`javac FileName.java` → this is mainly used to compile the java program

`java FileName` → this is mainly used to run the java program



.class

Understanding Structure of JAVA Program

name of
class

Main.java

```
public class Main {  
    public static void main(String[] args) {  
        System.out.println("Codegreedy Batch 6");  
    }  
}
```

Main. main

```
class PrintStream
```

```
}
```

```
    public void println()
```

```
    {
```

```
    }
```

```
}
```

· System.out.println()

out.println()

PrintStream out

↓ ↓ Input
output

System OS

```
class PrintStream
```

```
}
```

```
public void println()
```

```
{
```

```
}
```

```
}
```

```
class System
```

```
}
```

```
PrintStream out
```

```
}
```

```
System.out.println()
```

class Box

} fun()

Box.fun()

{

Box obj

obj.x

obj.fun()

```
void add()  
{  
}
```

```
{  
}
```

```
main()  
{  
}
```

```
, add()  
}
```

How to Print in JAVA Programming

Variables in JAVA

Data Type in JAVA

